



UNIVERSITY OF MORATUWA
SRI LANKA

Design of Local Area Network

Submitted by

ABEYWARDHANA T.C.W.

Registration No: 230012U

Peiris E.A.S.S.

Registration No: 230469B

Maldeniya P.A.D.G.R.

Registration No: 230390A

Ranainghe A.G.N.S

Registration No: 230520B

Department of Electronic and Telecommunication Engineering
University of Moratuwa

April 3, 2026

Contents

1	Introduction	2
2	Design Requirements and Assumptions	2
2.1	Scalability Considerations	2
2.2	Redundancy Requirements	2
2.3	Cost Considerations	3
3	University Backbone Network Design	3
3.1	Design Approach	4
3.2	Network Topology	4
3.3	IP Addressing Scheme	5
3.3.1	IPv4 Addressing Plan	5
3.3.2	IPv6 Addressing Plan	6
3.3.3	VLAN Allocation and Gateway Plan	7
3.4	Routing Strategy	7
3.5	Backbone Network Simulation	8
3.5.1	DHCP and IPv6 Configuration	8
3.5.2	Connectivity Verification	9
4	ENTC Local Area Network Design	10
4.1	Overview of Existing LAN	10
4.2	ENTC Network Design	11
4.3	ENTC Network Simulation Results	11
4.3.1	DHCP and IPv6 Configuration	11
4.3.2	Connectivity Verification	12
5	Component Selection and Justification	13
5.1	Active Network Components	14
5.1.1	Cisco Catalyst 3650-24PS Multilayer Switch	14
5.1.2	Access Switches	14
5.1.3	Wireless Access Points	14
5.1.4	Routers	15
5.1.5	Servers	15
5.2	Passive Network Components	15
5.2.1	UTP Ethernet Cables	15
5.2.2	Fibre Optic Backbone Links	15
5.2.3	Patch Panels and Racks	16
5.3	Backbone Network Device Features Relevant to the Design	16
5.4	Bill of Quantities for UoM Backbone Network	17
5.5	Bill of Quantities for ENTC Network	17

1 Introduction

This project presents the design of a scalable and cost-effective backbone network for the University of Moratuwa together with the local area network (LAN) of the Department of Electronic and Telecommunication Engineering (ENTC) building. The proposed design is developed with attention to scalability, reliability, efficient addressing, and overall network performance.

The aim is to create a backbone structure that can support current traffic demands while also allowing future expansion with minimal redesign. To evaluate the practicality of the design, the network is simulated using Cisco Packet Tracer, allowing connectivity, routing behaviour, and general functionality to be verified before implementation.

2 Design Requirements and Assumptions

The backbone network and the ENTC building LAN are designed based on several practical requirements and assumptions. These are intended to ensure that the final network is reliable, manageable, scalable, and economically reasonable for a university environment.

2.1 Scalability Considerations

The design is expected to support both present users and future growth in the number of students, staff members, and connected devices. The following considerations are included:

- Adequate IP address space is allocated using IPv4 VLSM and IPv6 /64 prefixes to support departments and shared facilities efficiently.
- VLAN-based segmentation is used so that new departments, laboratories, or service units can be added without major reconfiguration.
- A hybrid backbone structure allows new nodes and links to be introduced more easily while maintaining connectivity.
- Access and distribution devices are selected with enough capacity to support future bandwidth growth.

2.2 Redundancy Requirements

High availability is important in a university network because many services depend on uninterrupted communication. To improve fault tolerance, the design includes:

- Multiple paths between backbone nodes so that traffic can be rerouted if a link or device fails.

- Redundant uplinks between core and distribution devices to reduce single points of failure.
- Spanning Tree Protocol (STP) to prevent switching loops in Layer 2 segments.
- Dynamic routing with OSPF to support efficient path selection and automatic recovery during failures.

2.3 Cost Considerations

The design also attempts to balance performance with cost. Since budget is always a practical concern, the following assumptions are made:

- Network devices are selected by considering both performance requirements and budget limitations.
- Structured cabling and shared backbone resources are used to reduce deployment cost.
- Passive components such as cables, racks, and patch panels are chosen based on durability and cost-effectiveness.

3 University Backbone Network Design

The university backbone network acts as the central communication path that interconnects faculties, departments, and common service areas across the campus. A strong backbone is essential because it supports high-speed data transfer, service availability, and reliable communication between different parts of the university.

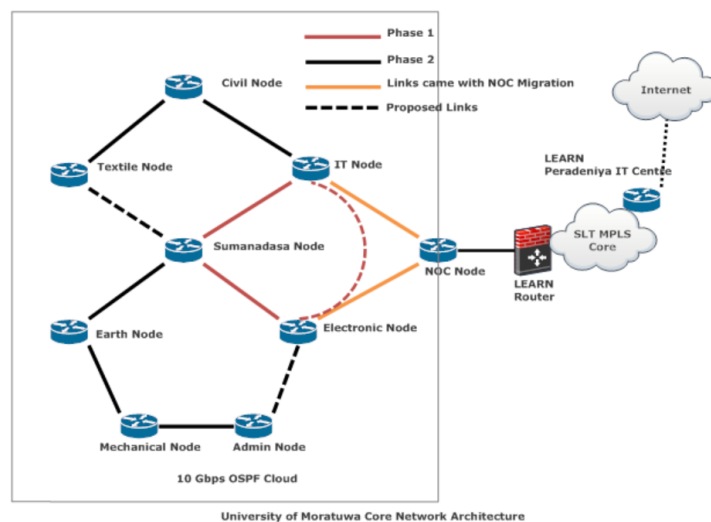


Figure 1: University of Moratuwa core network architecture.

Source: <https://uom.lk/cites/services/manage-backbone-network>

3.1 Design Approach

A hierarchical network design is adopted to improve scalability, simplify management, and support better performance. In this approach, the network is divided into three main layers:

- **Core Layer:** This is the high-speed backbone of the entire university network. It is responsible for transporting traffic quickly and reliably between major sections of the network.
- **Distribution Layer:** This layer connects individual faculties, departments, and buildings to the core. It also handles routing, policy control, and inter-VLAN communication.
- **Access Layer:** This is the layer where end-user devices such as computers, printers, and wireless access points connect to the network. It provides user access and supports VLAN-based segmentation.

This layered approach improves scalability, makes troubleshooting easier, and allows future expansion without requiring a full redesign of the network.

3.2 Network Topology

The proposed university backbone follows a hybrid topology that combines features of both ring and star structures. This approach is selected to improve reliability and maintain efficient communication across the campus.

- Backbone devices are connected in a way that provides continuous paths between important nodes.
- Additional links are introduced between selected nodes to improve resilience and provide alternate routes.
- As a result, each major node can reach others through more than one possible path.

This hybrid design offers several practical advantages:

- **Redundancy:** A single link failure is less likely to interrupt communication.
- **Fault Tolerance:** Traffic can be redirected through alternate paths when faults occur.
- **Improved Performance:** Traffic can be distributed more effectively across multiple links.
- **Scalability:** New nodes and additional links can be introduced without significantly affecting the existing structure.

Therefore, the hybrid backbone topology provides a good balance between performance, reliability, and cost for a large campus network.

3.3 IP Addressing Scheme

A structured IP addressing scheme is developed for the university network to support scalability, efficient subnetting, and easier network management. The design uses Variable Length Subnet Masking (VLSM) so that each department or service unit receives an address block that matches its expected number of users while minimizing address wastage.

3.3.1 IPv4 Addressing Plan

For IPv4, the university LAN is organized by assigning separate subnets to departments, faculties, and common service areas. Larger units are allocated larger subnet blocks, while smaller units are assigned smaller subnet sizes. This approach improves address utilization and also simplifies routing, VLAN mapping, and future expansion.

Department / Unit	IP Allocation	No. of Bits
Faculty of Engineering (FoE)		
CE	1022	10
EE	1022	10
ENTC	1022	10
ME	1022	10
MT	510	9
CH	1022	10
CS	2046	11
ER	1022	10
EM	1022	10
TMLE	1022	10
TEXT	1022	10
Other Faculties		
FoIT	4094	12
ARCHI	4094	12
Business	1022	10
MED	1022	10
Common Facilities		
Library	2046	11
CENTRA	2046	11
WALA	1022	10
MATH DEPT	1022	10
ADMIN	1022	10

Table 1: Estimated users and IP address allocation for the university LAN design

- Medium-sized departments are assigned /22 subnets.
- Smaller units are assigned /23 subnets.

Each department is mapped to a unique VLAN. This improves network segmentation, enhances security, and makes traffic management more efficient across the campus network.

3.3.2 IPv6 Addressing Plan

An IPv6 addressing scheme is introduced to support long-term scalability and modern networking requirements. For the purpose of simulation and documentation, a global unicast prefix is assigned to the university network.

- **IPv6 Prefix:** 2001:db8:acad::/48

IPv6 addresses are assigned using Stateless Address Autoconfiguration (SLAAC). With this method, hosts automatically generate their own IPv6 addresses by combining the advertised network prefix with an interface identifier.

- **Address Assignment Method:** SLAAC

The Layer 3 switch sends Router Advertisements (RA) to inform devices about the network prefix and the default gateway. Using these advertisements, hosts can configure their own IPv6 addresses without requiring a separate DHCPv6 server.

- **Default Gateway:** Link-local address of the VLAN interface (learned via RA)

The use of SLAAC provides several advantages:

- Automatic IP address configuration without manual setup
- No requirement for a DHCP server
- Simplified network management
- Efficient support for a large number of devices

A dual-stack approach using both IPv4 and IPv6 is adopted to maintain compatibility with existing IPv4-based systems while enabling IPv6 communication for future growth.

3.3.3 VLAN Allocation and Gateway Plan

The VLAN structure is designed so that each department or unit is assigned a dedicated VLAN, a default gateway at the primary core, and a corresponding subnet range. This makes the network easier to manage and supports better isolation between departments.

VLAN ID	Department Name	Gateway (Primary Core)	Subnet Mask	Usable IP Range
10	ARCHI	163.168.0.1	255.255.240.0	163.168.0.2 – 163.168.15.254
20	FoIT	163.168.16.1	255.255.248.0	163.168.16.2 – 163.168.23.254
30	CS	163.168.24.1	255.255.248.0	163.168.24.2 – 163.168.31.254
40	LIB	163.168.32.1	255.255.248.0	163.168.32.2 – 163.168.39.254
50	CENTRA	163.168.48.1	255.255.248.0	163.168.48.2 – 163.168.55.254
60	ADMIN	163.168.40.1	255.255.252.0	163.168.40.2 – 163.168.43.254
70	CE	163.168.56.1	255.255.252.0	163.168.56.2 – 163.168.59.254
80	ENTC	163.168.44.1	255.255.252.0	163.168.44.2 – 163.168.47.254
90	EE	163.168.60.1	255.255.252.0	163.168.60.2 – 163.168.63.254
100	ME	163.168.64.1	255.255.252.0	163.168.64.2 – 163.168.67.254
110	BUSI	163.168.68.1	255.255.252.0	163.168.68.2 – 163.168.71.254
120	MED	163.168.72.1	255.255.252.0	163.168.72.2 – 163.168.75.254
130	WALA	163.168.76.1	255.255.252.0	163.168.76.2 – 163.168.79.254
140	MATH DEPT	163.168.80.1	255.255.252.0	163.168.80.2 – 163.168.83.254
150	CH	163.168.84.1	255.255.252.0	163.168.84.2 – 163.168.87.254
160	ER	163.168.88.1	255.255.252.0	163.168.88.2 – 163.168.91.254
170	EM	163.168.92.1	255.255.252.0	163.168.92.2 – 163.168.95.254
180	TEXT	163.168.96.1	255.255.252.0	163.168.96.2 – 163.168.99.254
190	TMLE	163.168.100.1	255.255.252.0	163.168.100.2 – 163.168.103.254
200	MT	163.168.104.1	255.255.254.0	163.168.104.2 – 163.168.105.254

Table 2: VLAN allocation and IPv4 gateway configuration for university departments

3.4 Routing Strategy

The routing strategy for the University of Moratuwa backbone and the ENTC building LAN is designed to ensure efficient, scalable, and reliable data delivery across all departments.

- **Backbone Routing (Cisco Catalyst 3650 Switches):** Open Shortest Path First (OSPF) is used as the dynamic routing protocol between the distribution and core switches. OSPF is selected because it supports:
 - Fast convergence and adaptive path selection
 - Scalability across multiple VLANs and departments
 - Support for multiple paths in the hybrid mesh-ring topology
- **Internal Department Routing:** Within each department such as ENTC, routing is handled using Shortest Path Tree (SPT). This helps to ensure:
 - Optimal routing between devices inside the department network

- Low-latency communication between PCs, servers, and local resources
- Predictable traffic paths that minimize loops and congestion

Summary: Using OSPF for inter-department communication and SPT for intra-department traffic provides a balanced solution with good scalability, efficient routing, and low operational complexity.

3.5 Backbone Network Simulation

To validate the proposed backbone design, the complete university backbone network was implemented and tested in Cisco Packet Tracer. The simulation was used to verify IP address assignment, IPv6 configuration, inter-VLAN communication, and overall backbone connectivity across the campus network.

3.5.1 DHCP and IPv6 Configuration

The following figures show sample end-device configurations obtained from the simulation. These results confirm that the devices successfully receive valid IPv4 and IPv6 parameters according to the addressing plan.

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: FE80::200:CFF:FED6:B830
    Link-local IPv6 Address . . . . .: 2001:DB8:ACAD:80:200:CFF:FED6:B830
    IPv6 Address . . . . .: 163.168.44.2
    IPv4 Address . . . . .: 255.255.252.0
    Subnet Mask . . . . .: FE80::202:16FF:FEE9:A208
    Default Gateway . . . . .: 163.168.44.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: 
    IPv6 Address . . . . .: 
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 
    0.0.0.0

C:\>
```

Figure 2: IP configuration of PC0 at ENTC

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: FE80::2D0:FFFF:FEB6:441
    Link-local IPv6 Address . . . . .: 2001:DB8:ACAD:60:2D0:FFFF:FEB6:441
    IPv6 Address . . . . .: 163.168.40.2
    IPv4 Address . . . . .: 255.255.255.0
    Subnet Mask . . . . .: FE80::202:16FF:FEE9:A206
    Default Gateway . . . . .: 163.168.40.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: 
    IPv6 Address . . . . .: 
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 
    0.0.0.0

C:\>
```

Figure 3: IP configuration of PC3 at ADMIN

The above configurations show that both IPv4 and IPv6 addressing are functioning correctly in the simulated environment. Devices are able to obtain the required network parameters, confirming that the DHCP and IPv6 configuration approach is operating as expected.

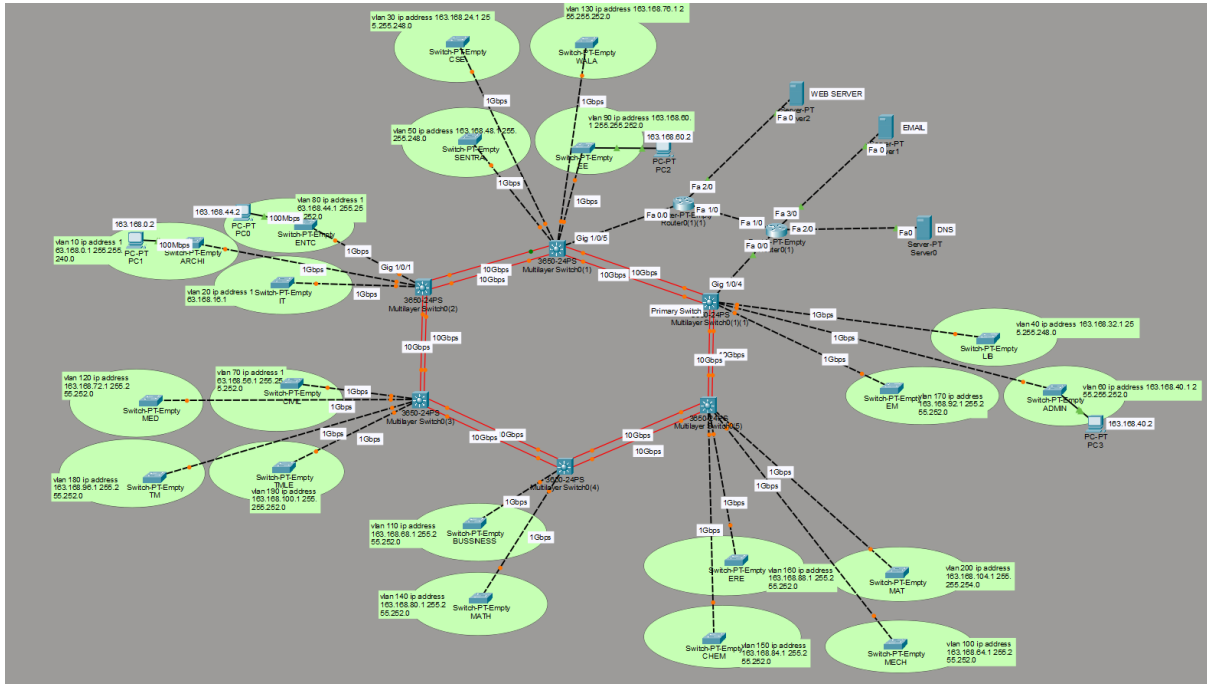


Figure 4: UoM backbone logical topology

The above figure illustrates the logical backbone topology of the University of Moratuwa network implemented in the simulation. It shows how the core, distribution, and access layer devices are interconnected to provide end-to-end communication across different faculties and units. The topology follows a hybrid mesh-ring structure, ensuring multiple redundant paths for data transmission. This design improves reliability, supports efficient routing using OSPF, and enables scalable expansion of the campus network.

3.5.2 Connectivity Verification

The connectivity of the proposed backbone network was verified using ping, traceroute, web access, and email communication tests in Cisco Packet Tracer.

```
C:\>ping 163.168.0.2

Pinging 163.168.0.2 with 32 bytes of data:

Reply from 163.168.0.2: bytes=32 time=202ms TTL=127
Reply from 163.168.0.2: bytes=32 time=31ms TTL=127
Reply from 163.168.0.2: bytes=32 time=183ms TTL=127
Reply from 163.168.0.2: bytes=32 time=18ms TTL=127

Ping statistics for 163.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 202ms, Average = 108ms

C:\>
```

Figure 5: Ping test between ENTIC and ARCHII

```
C:\>tracert 163.168.40.2

Tracing route to 163.168.40.2 over a maximum of 30 hops:

  0  109 ms  0 ms  0 ms  163.168.44.1
  1  37 ms   26 ms  39 ms  163.168.40.2

Trace complete.

C:\>
```

Figure 6: Traceroute from ENTIC to ADMIN

IPv4 Ping and Traceroute Tests These results confirm successful communication between different departments through the configured backbone network.

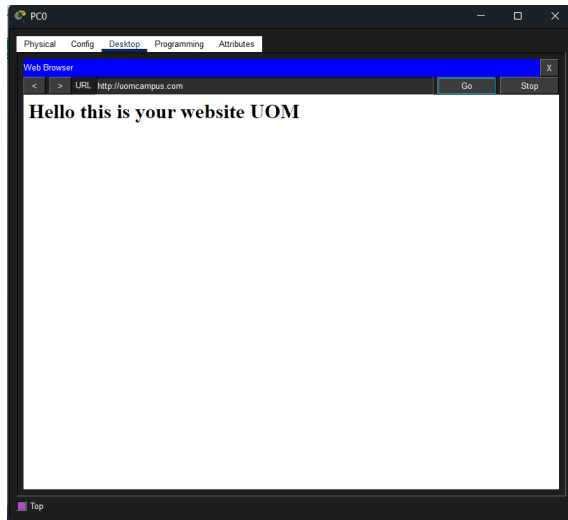


Figure 7: Accessing the university web page from a PC in ENTC

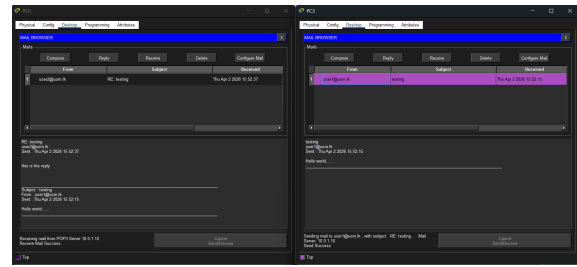


Figure 8: Email communication from ENTC to ADMIN

Application Layer Testing The successful results verify that both network connectivity and application services are functioning correctly in the simulation.

4 ENTC Local Area Network Design

4.1 Overview of Existing LAN

The existing ENTC LAN includes multiple laboratories, staff areas, and wireless access points distributed across different floors. The following figure shows the overall network structure currently used in the department.

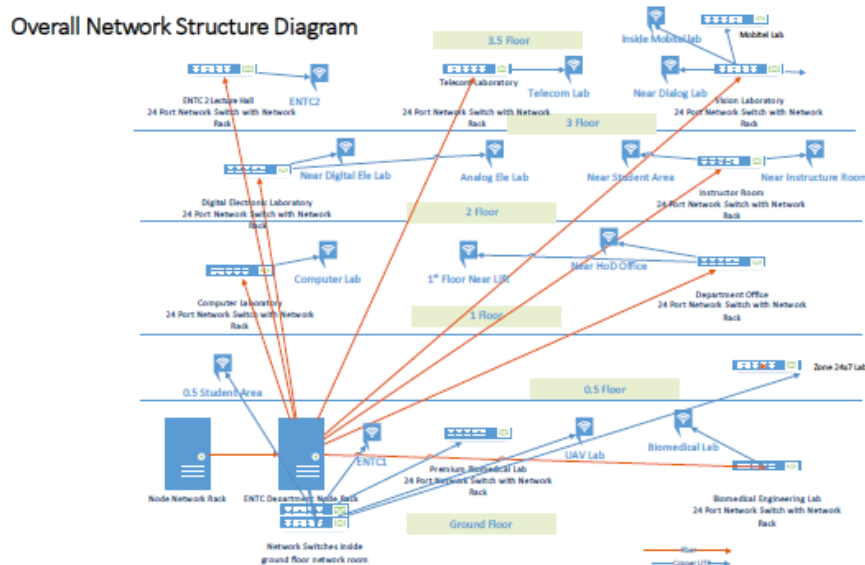


Figure 9: ENTC overall network structure diagram. ACK: Mr. Damith Kandage.

4.2 ENTC Network Design

Based on the existing LAN arrangement, a logical network architecture was developed for the simulation. It includes the department backbone, access switches, access points, and end devices connected across the different floors.

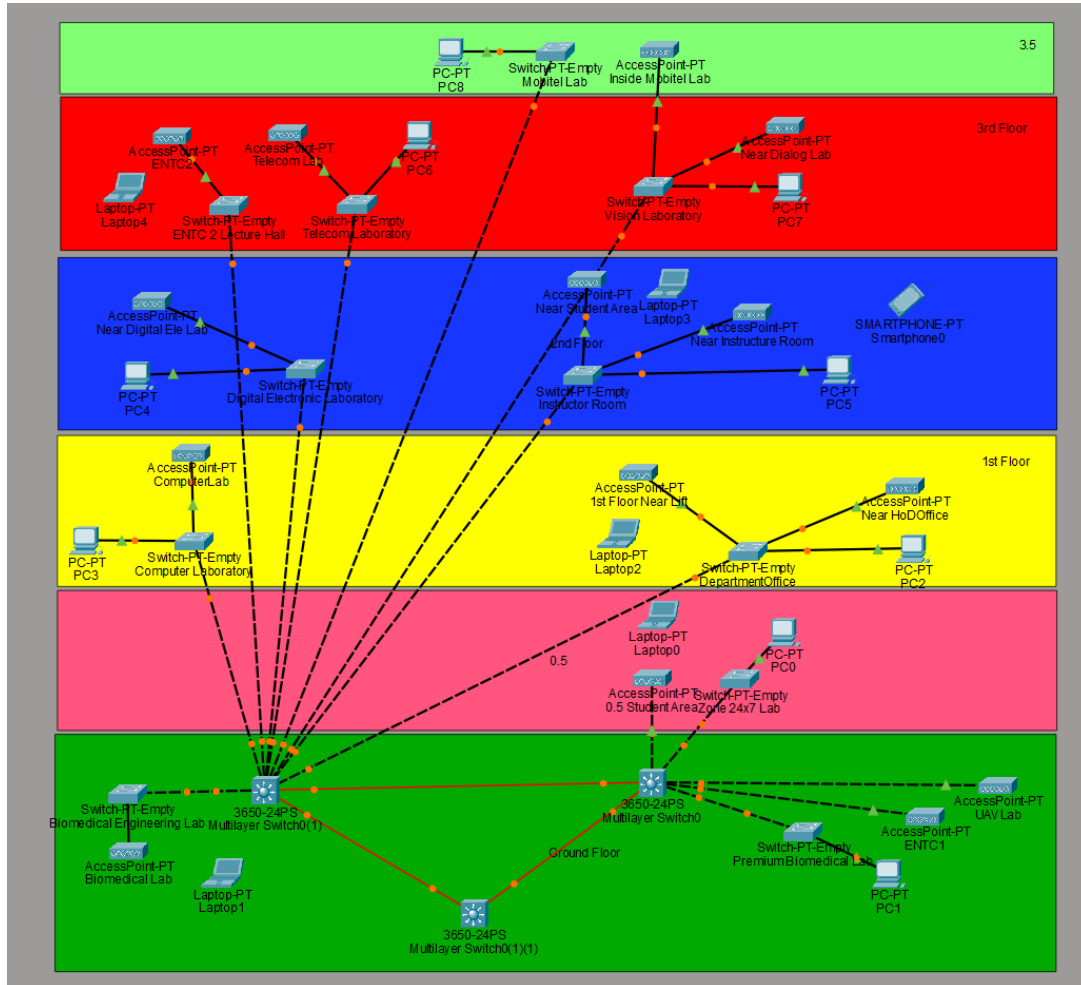


Figure 10: ENTC logical network architecture. Due to the use of Spanning Tree Protocol (STP), some redundant links are in the blocking state.

4.3 ENTC Network Simulation Results

The ENTC building network was simulated using Cisco Packet Tracer to verify the IP addressing scheme and communication within the department LAN.

4.3.1 DHCP and IPv6 Configuration

The simulation results confirm that the ENTC LAN devices receive valid IPv4 and IPv6 addresses based on the configured addressing plan.

- **IPv4 DHCP Subnet:** 163.168.1.0/24

- **Subnet Mask:** 255.255.255.0
- **IP Range for Hosts:** 163.168.1.2 to 163.168.1.254
- **Default Gateway:** 163.168.1.1
- **DNS Server:** 8.8.8.8
- **IPv6 Prefix:** 2001:DB8:ACAD:0::/64
- **IPv6 Address Assignment:** Stateless Address Autoconfiguration (SLAAC)
- **IPv6 Default Gateway:** FE80::240:BFF:FE9E:6D3E (VLAN interface link-local address)

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::240:BFF:FE9E:6D3E
    IPv6 Address . . . . .: 2001:DB8:ACAD:0:240:BFF:FE9E:6D3E
    IPv4 Address . . . . .: 163.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: FE80::290:21FF:FEBE:2DA4
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
                                0.0.0.0

C:\>
```

Figure 11: IP configuration of PC1 in the ENTC LAN

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:43FF:FE46:C071
    IPv6 Address . . . . .: 2001:DB8:ACAD:0:201:43FF:FE46:C071
    IPv4 Address . . . . .: 163.168.1.5
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: FE80::290:21FF:FEBE:2DA4
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:
                                0.0.0.0

C:\>
```

Figure 12: IP configuration of PC3 in the ENTC LAN

4.3.2 Connectivity Verification

Connectivity within the ENTC LAN was verified using ping, traceroute, and wireless connection tests in Cisco Packet Tracer.

```
C:\>ping 163.168.1.34

Pinging 163.168.1.34 with 32 bytes of data:

Reply from 163.168.1.34: bytes=32 time=58ms TTL=128
Reply from 163.168.1.34: bytes=32 time<1ms TTL=128
Reply from 163.168.1.34: bytes=32 time=24ms TTL=128
Reply from 163.168.1.34: bytes=32 time=26ms TTL=128

Ping statistics for 163.168.1.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 58ms, Average = 27ms

C:\>
```

Figure 13: Ping test between PC1 and Phone0

IPv4 Ping Test The successful replies confirm proper communication between devices in the ENTC network.

```

C:\>tracert 163.168.1.34

Tracing route to 163.168.1.34 over a maximum of 30 hops:

  1  25 ms   29 ms   18 ms   163.168.1.34

Trace complete.

C:\>

```

Figure 14: Traceroute between PC1 and Phone0

Traceroute Test The traceroute result confirms that packets reach the destination successfully through the configured network.

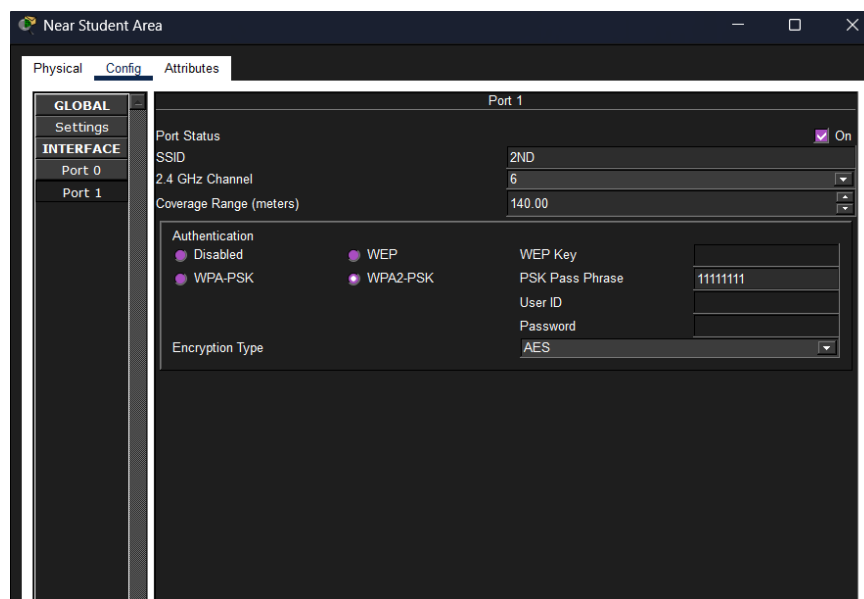


Figure 15: Wireless connection configuration

Wireless Connection This confirms that wireless devices can connect properly to the ENTC LAN.

5 Component Selection and Justification

The components used in this project were selected according to the requirements of the proposed University of Moratuwa backbone network and the ENTC local area network. The design includes both active and passive network components. Active devices are responsible for switching, routing, wireless connectivity, and service delivery, while passive components provide the physical links required for communication.

5.1 Active Network Components

5.1.1 Cisco Catalyst 3650-24PS Multilayer Switch

The Cisco Catalyst 3650-24PS multilayer switch is used as the main Layer 3 switching device in both the university backbone and the ENTC LAN design.

Justification

- Supports both Layer 2 switching and Layer 3 routing functions.
- Used for inter-VLAN routing between different departments and units.
- Suitable for implementing OSPF in the university backbone.
- Supports high-speed uplinks required for the backbone connections.
- Acts as the main aggregation point for multiple access switches.

5.1.2 Access Switches

Access switches are used to connect end devices such as PCs, laptops, laboratory systems, and local department devices.

Justification

- Provide connectivity for end-user devices at the access layer.
- Allow VLAN-based separation between laboratories, offices, and other areas.
- Simplify network expansion by allowing additional hosts to be connected easily.
- Suitable for structured LAN deployment within the ENTC building.

5.1.3 Wireless Access Points

Wireless access points are included in the ENTC LAN to provide Wi-Fi connectivity for laptops and smartphones in different areas of the department.

Justification

- Enable wireless access for mobile users.
- Extend network availability to areas where wired access is less practical.
- Support the integration of wireless devices into the departmental LAN.
- Improve usability in student areas, laboratories, and office spaces.

5.1.4 Routers

Routers are used in the backbone design to connect the internal university network with external services and server networks.

Justification

- Provide routing between the campus backbone and service networks.
- Support packet forwarding toward DNS, web, and email servers.
- Help separate internal switching functions from external network access.

5.1.5 Servers

Servers are included in the design for DNS, web, and email services.

Justification

- DNS server enables name resolution within the network.
- Web server is used to verify application-layer web connectivity.
- Email server is used to test end-to-end communication services.
- These services help validate the correctness of the overall network design.

5.2 Passive Network Components

5.2.1 UTP Ethernet Cables

UTP Ethernet cables are used for host-to-switch and access-layer connections in the LAN.

Justification

- Suitable for short-distance communication within laboratories and offices.
- Cost-effective and widely used for LAN deployment.
- Easy to install and maintain.

5.2.2 Fibre Optic Backbone Links

Fiber links are considered for the backbone interconnections between the main switching nodes.

Justification

- Suitable for high-speed backbone communication.
- Support higher bandwidth than standard copper links.
- Appropriate for inter-building or inter-floor backbone connections.

5.2.3 Patch Panels and Racks

Patch panels and network racks are required for the practical implementation of the physical network.

Justification

- Help organize and manage structured cabling.
- Improve maintenance and troubleshooting.
- Provide proper mounting and protection for network devices.

5.3 Backbone Network Device Features Relevant to the Design

The devices selected in this project support the main technical requirements of the proposed network design:

- **VLAN Support:** Required for departmental network segmentation.
- **Inter-VLANs Routing:** Necessary for communication between different VLANs.
- **Dynamic Routing:** OSPF is used in the backbone for efficient path selection.
- **Wireless Connectivity:** Access points support access to mobile devices in the ENTC LAN.
- **Application Services:** DNS, web, and email servers confirm end-to-end service functionality.
- **Scalability:** The design allows additional switches, access points, and end devices to be added in the future.

5.4 Bill of Quantities for UoM Backbone Network

Name	Unit Price (USD)	Quantity	Sub Total (USD)
Cisco Catalyst 3650-24PS switch	309.99	6	1,859.94
Cisco Catalyst 2960L 24-Port GigE switch	1,937.00	20	38,740.00
Cisco 2911/K9 router	1,844.00	2	3,688.00
1U rack server (Lenovo ThinkSystem SR250 V3)	3,432.94	3	10,298.82
Outdoor armored single-mode OS2 fiber cable	2.657/m	2,500 m	6,643.70
12U wall-mount rack cabinet	547.12	6	3,282.72
24-port Cat6 patch panel	89.16	20	1,783.20
Other (network sockets, patch cords, connectors)	1,500.00	1 lot	1,500.00
Total			67,796.38

Table 3: Bill of quantities for the UoM backbone network

5.5 Bill of Quantities for ENTC Network

Name	Unit Price (USD)	Quantity	Sub Total (USD)
Cisco Catalyst 3650-24PS switch	309.99	3	929.97
Cisco Catalyst 2960L 24-Port GigE switch	1,937.00	11	21,307.00
Cisco Catalyst 9105AX indoor access point	336.00	14	4,704.00
Cat6 bulk cable, 1000 ft box	325.24	5	1,626.20
Indoor/outdoor armored OM3 multi-mode fiber cable	3.084/m	400 m	1,233.60
12U wall-mount rack cabinet	547.12	6	3,282.72
24-port Cat6 patch panel	89.16	11	980.76
Other (network sockets, patch cords, connectors)	600.00	1 lot	600.00
Total			34,664.25

Table 4: Bill of quantities for the ENTC network

Design Files

The design files related to this project can be accessed using the following link:

[Google Drive Folder – Design Files](#)